

An Improved Analysis of the Empirical Convergence Rate of DPO-Mix-R

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Abstract

This note gives an improved analysis of the empirical DPO-Mix-R analysis in the tabular bandit setting of *The Crucial Role of Samplers in Online Direct Preference Optimization*. In the exact-gradient setting, the DPO-Mix-R sampler cancels the linear term in the pairwise error update, leaving a quadratic one-step exact-gradient error. With empirical gradients, this quadratic map is perturbed by centered sub-Gaussian noise of scale σ . Under the same moment-stability input used in the empirical proof, we show that empirical DPO-Mix-R reaches the $O(\sigma)$ root-mean-square error floor after a deterministic horizon of order $O(\log \log(1/\sigma))$.

Summary of the refinement. The original empirical theorem in [1] proves that, at $T = \lfloor \log(1/\sigma) \rfloor$,

$$\sqrt{\mathbb{E} \delta(a, a'; \theta^{(T)})^2} \leq O(\sigma).$$

The observation here is that the transient is shorter. On a high-probability event the empirical dynamics obey

$$\|\delta_{t+1}\|_\infty \leq K \|\delta_t\|_\infty^2 + \varepsilon, \quad \varepsilon \asymp \sigma \sqrt{\log(1/\sigma)},$$

so the quadratic phase reaches the ε -window in $O(\log \log(1/\sigma))$ steps. One final fresh empirical update then gives an L^2 contribution of order σ^2 , while the deterministic quadratic remainder is only $O(\varepsilon^4)$. The complement of the good event is absorbed by choosing its probability to be of order σ^4 and using a fourth-moment stability bound.

1 Setup and notation

Let $\mathcal{Y}$ be a finite action set and set $A = |\mathcal{Y}|$. Rewards are normalized as

$$r(a) \in [0, 1], \quad a \in \mathcal{Y}.$$

For a tabular policy π_θ , define the relative logit

$$q_\theta(a) := \log \frac{\pi_\theta(a)}{\pi_{\text{ref}}(a)}.$$

The pairwise error studied in the paper is

$$\delta(a, a'; \theta) := r(a) - r(a') - \beta(q_\theta(a) - q_\theta(a')). \quad (1)$$

Equivalently, under the common normalization $\pi_{\text{ref}} = \pi_{\theta(0)}$ and centered logits, this is

$$\delta(a, a'; \theta) = r(a) - r(a') - \beta(\theta_a - \theta_{a'}).$$

We also use

$$\Delta(a, a'; \theta) := \sigma(r(a) - r(a')) - \sigma(\beta(q_\theta(a) - q_\theta(a'))), \quad (2)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$.

Let

$$I := \mathcal{Y} \times \mathcal{Y}, \quad P := |I| \leq A^2.$$

For $i = (a, a') \in I$, write the time- t pairwise error as

$$\delta_t(i) := \Delta(a, a'; \theta^{(t)}), \quad \|\delta_t\|_\infty := \max_{i \in I} |\delta_t(i)|.$$

The initialization is $\pi_{\theta(0)} = \pi_{\text{ref}}$, so $\|\delta_0\|_\infty \leq 1$. Let $\mathcal{F}_t$ be the filtration generated by all empirical-gradient randomness up to time $t - 1$; then $\theta^{(t)}$ and δ_t are $\mathcal{F}_t$ -measurable.

2 One exact-gradient step for DPO-Mix-R

DPO-Mix-R uses two components:

$$(i) \pi^{s1} = \pi^{s2} = \text{Unif}(\mathcal{Y}), \quad (ii) \pi^{s1}(a) \propto e^{r(a)}, \quad \pi^{s2}(a) \propto e^{-r(a)}.$$

With the sampling coefficients in the paper, these two parts implement the factor

$$\frac{1}{\sigma'(r(a) - r(b))} = 2 + e^{r(a) - r(b)} + e^{-(r(a) - r(b))}.$$

Thus the mixed exact gradient has the same reweighted form as in Appendix A.2 of the paper. Taylor expansion at the reward difference gives

$$\Delta(a, b; \theta) = \sigma'(r(a) - r(b))\delta(a, b; \theta) - \frac{\sigma''(\xi_R(a, b; \theta))}{2}\delta(a, b; \theta)^2, \quad (3)$$

where $\xi_R(a, b; \theta)$ lies between $r(a) - r(b)$ and $\beta(q_\theta(a) - q_\theta(b))$.

Since $r(a) - r(b) \in [-1, 1]$, define

$$K := \frac{\sup_{x \in \mathbb{R}} |\sigma''(x)|}{\inf_{|x| \leq 1} \sigma'(x)} = \frac{1/(6\sqrt{3})}{\sigma'(1)} < \frac{1}{2}. \quad (4)$$

The precise numerical value is unimportant; what matters is that K is an absolute constant smaller than $1/2$.

Lemma 1 (Exact quadratic exact-gradient). *For the exact-gradient DPO-Mix-R update with learning rate $\eta = 1/(\beta^2 A)$, define $\theta_{\text{ex}}^{(t+1)}$ to be the parameter obtained by applying one exact-gradient DPO-Mix-R step from the current empirical iterate $\theta^{(t)}$, with the same learning rate $\eta = 1/(\beta^2 A)$. Define the corresponding one-step exact-gradient pairwise error by*

$$\delta_{t+1}^{\text{ex}}(i) := \Delta(i; \theta_{\text{ex}}^{(t+1)}), \quad i \in I.$$

Then

$$|\delta_{t+1}^{\text{ex}}(i)| \leq K \|\delta_t\|_\infty^2, \quad i \in I. \quad (5)$$

Proof. By the DPO-Mix-R reweighting and (3), the linear term in δ is exactly canceled by the learning rate $\eta = 1/(\beta^2 A)$. For $i = (a, a')$, the one-step exact-gradient pairwise error has the form

$$\delta_{t+1}^{\text{ex}}(a, a') = \frac{1}{2A} \sum_{b \in \mathcal{Y}} \left(\frac{\sigma''(\xi_R(a, b; \theta^{(t)}))}{\sigma'(r(a) - r(b))} \delta_t(a, b)^2 - \frac{\sigma''(\xi_R(a', b; \theta^{(t)}))}{\sigma'(r(a') - r(b))} \delta_t(a', b)^2 \right).$$

Using (4) and $|\delta_t| \leq \|\delta_t\|_\infty$ gives

$$|\delta_{t+1}^{\text{ex}}(a, a')| \leq \frac{K}{2A} \sum_{b \in \mathcal{Y}} (\delta_t(a, b)^2 + \delta_t(a', b)^2) \leq K \|\delta_t\|_\infty^2.$$

□

3 Empirical noise and moment stability

In empirical DPO, the exact gradient is replaced by an unbiased empirical gradient $G^{(t)}$. The centered coordinate error is assumed to satisfy the empirical-gradient condition from the paper: conditionally on $\mathcal{F}_t$,

$$\frac{G_a^{(t)} - \mathbb{E}[G_a^{(t)} | \mathcal{F}_t]}{\beta A} \text{ is sub-Gaussian with proxy variance } \sigma^2. \quad (6)$$

For $i = (a, a')$, define the induced pairwise noise

$$\nu_t(i) := \frac{(G_a^{(t)} - G_{a'}^{(t)}) - (\mathbb{E}[G_a^{(t)} | \mathcal{F}_t] - \mathbb{E}[G_{a'}^{(t)} | \mathcal{F}_t])}{\beta A}. \quad (7)$$

Then the empirical update decomposes into the one-step exact-gradient error plus the centered empirical-gradient noise:

$$\delta_{t+1}(i) = \delta_{t+1}^{\text{ex}}(i) + \nu_t(i), \quad i \in I. \quad (8)$$

Moreover, by standard closure properties of sub-Gaussian variables, there are absolute constants $c_0, c_2 > 0$ such that, for every $u > 0$,

$$\mathbb{E}[\nu_t(i) | \mathcal{F}_t] = 0, \quad (9)$$

$$\mathbb{P}(|\nu_t(i)| > u | \mathcal{F}_t) \leq 2 \exp \left(-\frac{u^2}{c_0 \sigma^2} \right), \quad (10)$$

$$\mathbb{E}[\nu_t(i)^2 | \mathcal{F}_t] \leq c_2 \sigma^2. \quad (11)$$

The high-probability argument controls the dynamics only on a good event. To convert it into an unconditional root-mean-square statement, we need the following fourth-moment stability input.

Assumption 1 (Fourth-moment stability). *For the deterministic horizon H under consideration, there exists a constant $B_4 < \infty$, independent of σ , such that*

$$\max_{0 \leq t \leq H} \max_{i \in I} \mathbb{E}|\delta_t(i)|^4 \leq B_4^4. \quad (12)$$

Remark 1 (Where Assumption 1 comes from). This is the same stability input supplied by the L^{2n} moment induction in the empirical proof of the paper. In that proof, one obtains bounds of the form

$$\mathbb{E}|\delta_t(i)|^{2n} \leq (12\sqrt{n}\sigma + 2^{-t})^{2n}$$

whenever $n2^t \lesssim 1/\sigma$. Taking $n = 2$ gives a uniform fourth-moment bound for every horizon $H = O(\log \log(1/\sigma))$. The analysis below uses only the consequence (12), so the proof is separated from the original moment induction.

4 Good event and log-log hitting time

Fix a deterministic horizon $H \geq 1$ and a failure budget $p \in (0, 1)$. Define

$$\rho := \sqrt{c_0 \log \frac{2PH}{p}}, \quad \varepsilon := \rho\sigma, \quad (13)$$

and the good event

$$\mathcal{G} := \{|\nu_t(i)| \leq \varepsilon \text{ for all } 0 \leq t < H, i \in I\}. \quad (14)$$

Lemma 2 (Good-event recursion). *Under (10), $\mathbb{P}(\mathcal{G}) \geq 1 - p$. Moreover, on $\mathcal{G}$,*

$$\|\delta_{t+1}\|_\infty \leq K\|\delta_t\|_\infty^2 + \varepsilon, \quad 0 \leq t < H. \quad (15)$$

Proof. For each fixed (t, i) , (10) and (13) give

$$\mathbb{P}(|\nu_t(i)| > \varepsilon \mid \mathcal{F}_t) \leq 2 \exp(-\rho^2/c_0) = \frac{p}{PH}.$$

A union bound over H times and P ordered pairs gives $\mathbb{P}(\mathcal{G}^c) \leq p$. On $\mathcal{G}$, combine (5) and (8), then maximize over i . $\square$

Set

$$B := \sqrt{\frac{\varepsilon}{K}}. \quad (16)$$

On $\mathcal{G}$, define the hitting time

$$\tau := \inf\{t \geq 0 : \|\delta_t\|_\infty < B\}.$$

Lemma 3 (Log-log hitting time). *Assume $\|\delta_0\|_\infty \leq 1$ and $2\sqrt{K\varepsilon} < 1$. On $\mathcal{G}$,*

$$\tau \leq L(H, p, \sigma) := \left\lceil \log_2 \left(\frac{\log(1/(2\sqrt{K\varepsilon}))}{\log(1/(2K))} \right) \right\rceil. \quad (17)$$

Consequently, if $H \geq L(H, p, \sigma) + 2$, then $\tau + 2 \leq H$ on $\mathcal{G}$.

Proof. As long as $\|\delta_t\|_\infty \geq B$, the noise term is dominated:

$$\varepsilon \leq K\|\delta_t\|_\infty^2.$$

Thus (15) gives $\|\delta_{t+1}\|_\infty \leq 2K\|\delta_t\|_\infty^2$. Let $z_t := 2K\|\delta_t\|_\infty$. During this phase,

$$z_{t+1} = 2K\|\delta_{t+1}\|_\infty \leq (2K\|\delta_t\|_\infty)^2 = z_t^2.$$

Since $z_0 \leq 2K < 1$, induction gives $z_t \leq (2K)^{2^t}$ before the hitting time. The inequality $\|\delta_t\|_\infty < B$ is equivalent to $z_t < 2\sqrt{K\varepsilon}$. Hence it is enough to have

$$(2K)^{2^t} \leq 2\sqrt{K\varepsilon},$$

which is exactly (17). $\square$

5 Reaching the RMS noise floor

The key point is that the second-moment estimate is taken after the good-event pathwise descent. We do not use an unconditional recursion and then try to replace the sup-norm fourth moment by pairwise second moments.

Lemma 4 (Two-step descent to the empirical floor). *Assume $H \geq \tau + 2$ on $\mathcal{G}$ and $4K\varepsilon^2 \leq \varepsilon$. Then, on $\mathcal{G}$,*

$$\|\delta_t\|_\infty \leq 2\varepsilon, \quad t = \tau + 1, \dots, H. \quad (18)$$

Furthermore, for every ordered pair $i \in I$,

$$\mathbb{E}[\delta_H(i)^2 \mathbf{1}_{\mathcal{G}}] \leq 32K^2\varepsilon^4 + 2c_2\sigma^2. \quad (19)$$

Proof. At the hitting time, $\|\delta_\tau\|_\infty < B$, and hence

$$\|\delta_{\tau+1}\|_\infty \leq KB^2 + \varepsilon = 2\varepsilon.$$

If $\|\delta_t\|_\infty \leq 2\varepsilon$ for some $t \geq \tau + 1$, then

$$\|\delta_{t+1}\|_\infty \leq K(2\varepsilon)^2 + \varepsilon = 4K\varepsilon^2 + \varepsilon \leq 2\varepsilon.$$

This proves (18). Since $H \geq \tau + 2$, the lock-in bound holds at time $H - 1$. Therefore, on $\mathcal{G}$,

$$|\delta_H^{\text{ex}}(i)| \leq K\|\delta_{H-1}\|_\infty^2 \leq 4K\varepsilon^2.$$

Using $\delta_H(i) = \delta_H^{\text{ex}}(i) + \nu_{H-1}(i)$ and $(x + y)^2 \leq 2x^2 + 2y^2$,

$$\delta_H(i)^2 \mathbf{1}_{\mathcal{G}} \leq 2(4K\varepsilon^2)^2 \mathbf{1}_{\mathcal{G}} + 2\nu_{H-1}(i)^2 \mathbf{1}_{\mathcal{G}}.$$

Taking expectations and applying (11) gives (19). $\square$

Lemma 5 (Bad-event absorption). *Under Assumption 1, for every ordered pair $i \in I$,*

$$\mathbb{E}[\delta_H(i)^2 \mathbf{1}_{\mathcal{G}^c}] \leq B_4^2 \sqrt{p}. \quad (20)$$

Proof. By Cauchy-Schwarz and Assumption 1,

$$\mathbb{E}[\delta_H(i)^2 \mathbf{1}_{\mathcal{G}^c}] \leq (\mathbb{E}|\delta_H(i)|^4)^{1/2} \mathbb{P}(\mathcal{G}^c)^{1/2} \leq B_4^2 \sqrt{p}.$$

$\square$

6 Main theorem

Theorem 1 (Log-log transient for empirical DPO-Mix-R). *Assume the empirical DPO-Mix-R update satisfies (8), the quadratic exact-gradient bound (??), the noise conditions (9)–(11), and Assumption 1 up to a deterministic horizon H . Suppose $\|\delta_0\|_\infty \leq 1$ and $\sigma < B_4$. Set*

$$p := \frac{\sigma^4}{B_4^4}, \quad \rho := \sqrt{c_0 \log \frac{2PH}{p}}, \quad \varepsilon := \rho\sigma. \quad (21)$$

If

$$H \geq L(H, p, \sigma) + 2, \quad 2\sqrt{K\varepsilon} < 1, \quad 4K\varepsilon^2 \leq \varepsilon, \quad (22)$$

then for every $a, a' \in \mathcal{Y}$,

$$\sqrt{\mathbb{E}[\delta(a, a'; \theta^{(H)})^2]} \leq C\sigma, \quad (23)$$

where one may take

$$C^2 = 2c_2 + 1 + 32K^2\rho^4\sigma^2. \quad (24)$$

In particular, along the deterministic log-log horizons in Corollary 1, the last term in (24) is $o(1)$.

Proof. By Lemma 3, the first condition in (22) implies $\tau + 2 \leq H$ on $\mathcal{G}$. Hence Lemma 4 gives

$$\mathbb{E}[\delta_H(i)^2 \mathbf{1}_{\mathcal{G}}] \leq 32K^2\varepsilon^4 + 2c_2\sigma^2 = (32K^2\rho^4\sigma^2 + 2c_2)\sigma^2.$$

By Lemma 5 and the choice of p ,

$$\mathbb{E}[\delta_H(i)^2 \mathbf{1}_{\mathcal{G}^c}] \leq B_4^2 \sqrt{\sigma^4/B_4^4} = \sigma^2.$$

Adding the good-event and bad-event contributions yields

$$\mathbb{E}[\delta_H(i)^2] \leq (32K^2\rho^4\sigma^2 + 2c_2 + 1)\sigma^2.$$

Taking square roots proves the claim. $\square$

Corollary 1 (Existence of an $O(\log \log(1/\sigma))$ deterministic horizon). *Fix finite A and constants K, c_0, c_2, B_4 . There exist constants $C_H, \sigma_0 > 0$, depending only on these quantities, such that for every $0 < \sigma \leq \sigma_0$, the horizon*

$$H = \left\lceil C_H \log \log \frac{e^e}{\sigma} \right\rceil \quad (25)$$

satisfies the hypotheses of Theorem 1. Consequently, for every $a, a' \in \mathcal{Y}$,

$$\sqrt{\mathbb{E}[\delta(a, a'; \theta^{(H)})^2]} \leq C\sigma.$$

Proof. With $p = \sigma^4/B_4^4$,

$$\rho^2 = c_0 \log \frac{2PHB_4^4}{\sigma^4} = O\left(\log \frac{1}{\sigma} + \log H\right) = O\left(\log \frac{1}{\sigma}\right)$$

for $H = O(\log \log(1/\sigma))$. Thus

$$\varepsilon = \rho\sigma = O\left(\sigma \sqrt{\log \frac{1}{\sigma}}\right) \rightarrow 0.$$

The two smallness conditions in (22) therefore hold for all sufficiently small σ . Moreover,

$$L(H, p, \sigma) = O\left(\log \log \frac{1}{\sigma}\right).$$

Choosing C_H larger than the implicit constant ensures $H \geq L(H, p, \sigma) + 2$ for all sufficiently small σ . Finally, the fourth-moment stability input holds for such horizons by the original moment induction, since $H = O(\log \log(1/\sigma))$ is much smaller than the range allowed by $n2^H \lesssim 1/\sigma$ with $n = 2$. The result follows from Theorem 1. $\square$

References

- [1] Ruizhe Shi, Runlong Zhou, and Simon S. Du. The crucial role of samplers in online direct preference optimization. *International Conference on Learning Representations*, 2025.